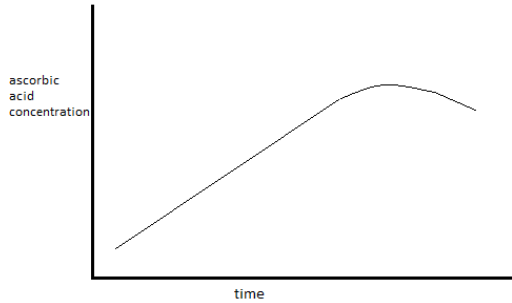


1(a)(i)	any one from: 1 <i>idea that</i> size (of apple) may not be, related / proportional, to ripeness ; 2 <i>idea that</i> (individual) apples may, ripen at different rates / AW ; 3 <i>idea that</i> other factors may affect, ripeness of apples / diameter of apples ;	1
1(a)(ii)	any one from: 1 (pick apples with the) same / similar / AW, sugar content / colour / softness / age ; 2 test / measure / AW, sugar content / colour / softness / age (of apples) ;	1
1(b)	1 correct axis labels ; 2 correct shaped line ;  The graph shows a line starting from the origin, rising linearly to a peak, and then slightly declining. The y-axis is labeled 'ascorbic acid concentration' and the x-axis is labeled 'time'.	2
1(c)(i)	1 90 ; 2 50 ;	2

Question	Answer	Marks												
1(c)(ii)	any nine from: - idea of test on at least 5 days ; - use apples from same, tree / variety / type ; - use, same / stated, storage conditions (of apples) ; prepare apple extract solutions: - remove the, skin / seeds, of the apple ; - use, set / stated, mass of apple ; - method to, grind / AW, apple ; - centrifuge / filter ; measure ascorbic acid concentration using DCPIP: - place, set / stated, volume of (apple extract) solution in, conical flask / AW ; - idea of adding DCPIP dropwise to (apple extract) solution until blue colour remains ; - shake / mix / AW (after each drop of DCPIP) ; - measure / record, volume of DCPIP used (to reach end point) ; - read volume from bottom of meniscus (in burette) ; - use at least three replicates for each day <u>and</u> calculate a mean (DCPIP volume) ; - safety comment with hazard <u>and</u> risk <u>and</u> precaution ; <table border="1"> <thead> <tr> <th>hazard</th><th>risk</th><th>precaution</th></tr> </thead> <tbody> <tr> <td>DCPIP</td><td>irritant / toxin / allergy</td><td>gloves / goggles / mask / PPE</td></tr> <tr> <td>apples</td><td>allergy</td><td>gloves / goggles / mask / PPE / medication / do not eat</td></tr> <tr> <td>knife / scalpel</td><td>injury</td><td>cut away from hand</td></tr> </tbody> </table>	hazard	risk	precaution	DCPIP	irritant / toxin / allergy	gloves / goggles / mask / PPE	apples	allergy	gloves / goggles / mask / PPE / medication / do not eat	knife / scalpel	injury	cut away from hand	9
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1(d)	any two from: - data is continuous ; - scatter diagram, suggests a linear, relationship ; - data is normally distributed ;	2												

Question	Answer	Marks
1(e)	any three from: for the conclusion: 1 apples have the second lowest β-carotene concentration (of the fruits shown) ; 2 quoted data from Fig. 1.3 to support ; 3 <i>idea that</i> spread of data for apple is lower (than most fruits that were tested, as SD is smaller) against the conclusion: 4 <i>idea that</i> need to calculate, SE / 95% CI (to make a conclusion) ; 5 no statistical test carried out ; 6 <i>idea that</i> only 5 fruits were tested ;	3

Question	Answer	Marks
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